
Joseph Kerry

West Bridgford, Nottingham • [GitHub](#) • [portfolio](#)

Employment

CoreFlow Operations Subcontractor (Feb 2026 – May 2026)

Part-Time Systems Developer/ Administrator

- Consulting documentation and requirements for SMEs.
- Developing systems from a user perspective to ensure usability and functionality.
- Partial development of Nottingham Forest Supporters Trust website.

Education

Master Of Science at NOTTINGHAM TRENT UNIVERSITY

MSc, Software Engineering

Current overall grade: Low Distinction (120/180 credits)

Major Project: Benchmarking Deep Learning Architectures for EEG Motor Imagery Classification on Open BCI's Galea Headset using Standardised Preprocessing Pipelines. (60 credits)

Bachelor Of Law with Honours at NOTTINGHAM TRENT UNIVERSITY

LLB Hons, Law with Business

Final Grade: Second Class Honours

Notable Projects

Convolutional Vision Transformer – Temporal Convolutional Network Hybrid Model

- Implemented a CVT-TCN using PyTorch for motor imagery classification (not for major project) on the EEGMMIDB dataset, converting raw time trials into time-frequency representations.
- Achieved a subject independent accuracy on 4 class motor imagery, achieving similar results to EEGNet with an end-to-end pipeline.

High Energy Physics Data Pipeline

- Python and FastAPI based data pipeline and analysis model for processing high-energy physics event data. Implemented async ingestion methods, event filtering, and statistical analysis endpoints.
- Designed a scalable REST API architecture with optimized query handling and structured data storage to handle high output datasets.

Machine Learning Library in C

- Implemented classical ML and DL algorithms in C, including supervised learning, clustering and neural network approaches which I applied for analysing particle simulation data produced by my high energy physics data pipeline project.

Distributed Cloud Load Balancer

- Built a Java based distributed cloud storage simulator across four dockerised server nodes, implemented client-side AES encryption, file chunking and load balanced distribution.
- Designed a dual database architecture with an MQTT based host manager supporting dynamic scaling.

Skills

Languages: Python, C#, C, Java

Databases: SQL, SQLite, PostgreSQL

Machine learning: PyTorch, NumPy, MNEPython, Pytest, Seaborn, Matplotlib

Other technical skills: Docker, Git